

Question

Conjugate base for Bronsted acids H_2O and HF are

- A** H_3O^+ and F^- , respectively
- B** OH^- and F^- , respectively
- C** H_3O^+ and H_2F^+ , respectively
- D** OH^- and H_2F^+ , respectively

Question

The conjugated acid of O^{2-} ion is-

- A** O^{2+}
- B** H^+
- C** H_3O^+
- D** OH^-

Question

In the reaction $\text{NH}_3 + \text{H}_2\text{O} \rightarrow \text{NH}_4^+ + \text{OH}^-$ water behaves as–

- A** Acid
- B** Base
- C** Neutral
- D** Both acid and Base

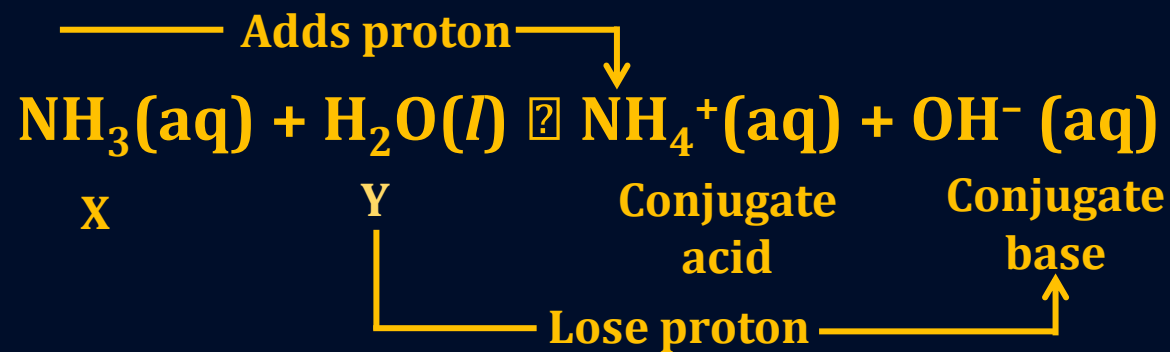
Question

The conjugated base of $(\text{CH}_3)_2\text{NH}_2^+$ is–

- A** CH_3NH_2
- B** $(\text{CH}_3)_2\text{N}^+$
- C** $(\text{CH}_3)_2\text{N}$
- D** $(\text{CH}_3)_2\text{NH}$

Question

In the reaction,



X and **Y** respectively are

A neutral and acid

B acid and base

C base and acid

D base and neutral

Question

Which of the following cannot act both as Bronsted acid and as Bronsted base?
[NEET (Odisha) 2019]



Question

Which of the following is least likely to behave as Lewis base?

[CBSE AIPMT 2011]



Question

Which equilibrium can be described as an acid-base reaction using the Lewis acid-base definition, but not using Bronsted-Lowry definition?

- A** $\text{NH}_3 + \text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{NH}_4^+$
- B** $\text{H}_2\text{O} + \text{CH}_3\text{COOH} \rightleftharpoons \text{H}_3\text{O}^+ + \text{CH}_3\text{COO}^-$
- C** $4\text{NH}_3 + [\text{Cu}(\text{H}_2\text{O})_4]^{2+} \rightleftharpoons [\text{Cu}(\text{NH}_3)_4]^{2+} + 4\text{H}_2\text{O}$
- D** $2\text{NH}_3 + \text{H}_2\text{SO}_4 \rightleftharpoons 2\text{NH}_4^+ + \text{SO}_4^{2-}$

Question

For which, dilution law is applicable?

- A** NaCl
- B** NaOH
- C** CH_3COONa
- D** CH_3COOH

Question

If $[H^+] = \frac{2}{3} \times 10^{-4}$ then find pH?

- A** 2.7
- B** 4.176
- C** 2
- D** 1

Question

If $\text{pH} = 3.31$, then find out $[\text{H}^+]$ (Approx)

- A** 3.39×10^{-4}
- B** 5×10^{-4}
- C** 3.0×10^{-3}
- D** None

Question

A beer has a pH of 4.30. What is the $[\text{H}_3\text{O}^+]$?

- A** 3.0×10^{-4}
- B** 2.0×10^{-4}
- C** 2.0×10^{-3}
- D** 5.0×10^{-5}

Question

The number of H^+ ions present in 1 mL of a solution having $\text{pH} = 13$ is:

- A** 10^{13}
- B** 6.023×10^{13}
- C** 6.023×10^7
- D** 6.023×10^{10}

Question

If $[\text{OH}^-] = 5.0 \times 10^{-5} \text{ M}$ then pH will be

- A** $5 - \log 5$
- B** $9 + \log 5$
- C** $\log 5 - 5$
- D** $\log 5 - 9$

Question

pOH of 0.002 M HNO_3 is:

- A** $11 + \log 2$
- B** $11 - \log 2$
- C** $-3 + \log 2$
- D** None of these

Question

The hydroxide ion concentration of a wine is 8×10^{-11} M. What is the pH of the wine?

- ☐ A 2.10
- ☐ B 2.9
- ☐ C 3.9
- ☐ D 4.9

Question

What is the pH of a 0.015 M $\text{Ba}(\text{OH})_2$ solution?

- A** 1.82
- B** 1.52
- C** 12.48
- D** 12.18

Question

For a 100 ml solution of 10^{-2} M NaOH the ratio pH : pOH would be-

- A 6 : 1
- B 1 : 6
- C 2 : 1
- D 10^{10} : 1

Question

The pH of a solution is 6.0. In this solution-

A $[H^+] = 100 [OH^-]$

B $[H^+] = 10 [OH^-]$

C $[H^+] = [OH^-]$

D $[H^+] = [OH^-]$

Question

Suppose at 60°C K_w of H_2O is 10^{-13} . Calculate the pH of H_2O and also define the pH scale.

Question

The unit of ionic product of water K_w are

A $\text{Mol}^{-1}\text{L}^{-1}$

B $\text{Mol}^{-2} \text{L}^{-2}$

C $\text{Mol}^{-2} \text{L}^{-1}$

D $\text{Mol}^2 \text{L}^{-2}$

Question

At 90°C, pure water has $[\text{H}_3\text{O}^+] = 10^{-6}$ mole litre⁻¹. What is the value of K_w at 90°C?

- A** 10^{-6}
- B** 10^{-12}
- C** 10^{-14}
- D** 10^{-8}

Question

The pK_w of water at 50°C is 13.40. An aqueous solution at 50°C has $\text{pH} = 7$. This solution is–

- A** Acidic
- B** Alkaline
- C** Neutral
- D** Amphoteric

Question

A solution has pOH equal to 13 at 298 K. The solution will be:

- A** highly acidic
- B** highly basic
- C** moderately basic
- D** unpredictable

Question

The pH of a 10^{-9} M solution of HCl in water is:

- A** 8
- B** -8
- C** Between 7 and 8
- D** Between 6 and 7

Question

An aqueous solution of HCl with pH 1.0 is diluted by adding equal volume of water (ignoring dissociation of water). The pH of HCl solution would (Given $\log 2 = 0.30$)

- A** reduce to 0.5
- B** increase to 1.3
- C** remain same
- D** increase to 2

Question

When the hydrogen ion concentration $[H^+]$ changes by a factor of 1000, the value of pH of the solution_____.

- A** increases by 1000 units
- B** decreases by 3 units
- C** decreases by 2 units
- D** increases by 2 units

Question

An analyst wants to convert. 1L HCl of pH = 1 to a solution of HCl of pH 2. The volume of water needed to do this dilution is _____ mL. (Nearest Integer)

[12 April, 2023 (Shift-I)]

Question

Equal volumes of three acid solutions of pH 3, 4 and 5 are mixed in a vessel. What will be the H^+ ion concentration in the mixture?

- A** $1.11 \times 10^{-4} \text{ M}$
- B** $3.7 \times 10^{-4} \text{ M}$
- C** $3.7 \times 10^{-3} \text{ M}$
- D** $1.11 \times 10^{-3} \text{ M}$

Question

If 100 mL of pH = 3 and 400 mL of pH = 3 is mixed, what will be the pH of the mixture

- A** 3.2
- B** 3.0
- C** 3.5
- D** 2.8

Question

The pH of the solution produced when an aqueous solution of strong acid pH 5 is mixed with equal volume of an aqueous solution of strong acid of pH 3 is:

- A** 3.3
- B** 3.5
- C** 4.5
- D** 4.0

Question

The pH of the solution containing 10 mL of a 0.1M NaOH and 10 mL of 0.05M H_2SO_4 would be

- ☐ A Zero
- ☐ B 1
- ☐ C >7
- ☐ D 7

Question

8 g NaOH and 4.9 g H_2SO_4 are present in one litre of the solution. What is its pH?

- A** 1
- B** 13
- C** 12
- D** 2

Question

The pH of the solution obtained by mixing 10 mL of 0.1 M HCl and 10 mL of 0.1 M NaOH is:

- ☐ A 8
- ☐ B 2
- ☐ C 7
- ☐ D None of these

Question

Following four solutions are prepared by mixing different volumes of NaOH and HCl of different concentrations, pH of which one of them will be equal to 1?



Question

Calculate the pH of solution formed by mixing of 300ml of 0.2 M NaOH, 200 ml of 0.4M H_2SO_4 and 500 mL of 0.1 M $\text{Ca}(\text{OH})_2$.

Question

pH of water is 7 at 25°C. If water is heated to 80°C., it's pH will:

- ☐ A Remains the same
- ☐ B Decrease
- ☐ C Increase
- ☐ D H^+ concentration increases, OH^- concentration decreases

Question

Which of the following mixing of 1 M base and 1 M acid leads to the largest increase in temperature?

- ☐ A 45 mL CH_3COOH and 25 mL NaOH
- ☐ B 50 mL HCl and 20 mL NaOH
- ☐ C 30 mL CH_3COOH and 30 mL NaOH
- ☐ D 30 mL HCl and 30 mL NaOH

Question

pH value of 0.001 M NaOH solution is_____.

[27 June, 2022 (Shift-II)]

For the following Assertion and Reason, the correct option is:

Assertion (A): The pH of water increase with increase in temperature.

Reason (R): The dissociation of water into H^+ and OH^- is an exothermic reaction

[8 Jan, 2020 (Shift-II)]

- A** Assertion is not true, but reason is true
- B** Both assertion and reason are false
- C** Both assertion and reason are true but the reason is not the correct explanation for the assertion
- D** Both assertion and reason are true, and the reason is the correct explanation for the assertion

Question

What is the pH of the resulting solution when equal volumes of 0.1 M NaOH and 0.01 M HCl are mixed?

- A** 7.0
- B** 1.04
- C** 12.65
- D** 2.0

Question

The pH of a mixture when a 50 mL solution of pH = 1 is mixed with a 50 mL of pH = 2 is:

- A** 0.56
- B** 1.26
- C** 1.76
- D** 2.06

Question

Which of the following is the weakest acid?

- A** Phenol ($K_a = 1.3 \times 10^{-10}$)
- B** Hydrocyanic acid ($K_a = 4.9 \times 10^{-10}$)
- C** Acetic acid ($K_a = 1.8 \times 10^{-5}$)
- D** Benzoic acid ($K_a = 6.5 \times 10^{-5}$)

Question

Which of the following is the strongest base?

- A** $\text{C}_6\text{H}_5\text{NH}_2$ ($\text{pK}_b = 9.42$)
- B** $\text{C}_6\text{H}_5\text{NHCH}_3$ ($\text{pK}_b = 9.15$)
- C** $\text{C}_6\text{H}_5\text{N}(\text{CH}_3)_2$ ($\text{pK}_b = 8.94$)
- D** $\text{C}_6\text{H}_5\text{NHC}_2\text{H}_5$ ($\text{pK}_b = 8.89$)

Question

Find out the value of α of 10^{-2} M HCN solution if $[H^+] = 10^{-3}$.

Question

For 10^{-3} M H_2CO_3 if $\alpha = 10\%$ then find out the value of pH?

Question

For 10 M CH_3COOH solution if $K_a = 10^{-5}$ then find out:

(i) α

(ii) $[\text{H}^+]$

(iii) pH

Question

A 0.2 molar solution of formic acid is 3.2% ionised. Its ionisation constant is

- A** 9.6×10^{-3}
- B** 2.1×10^{-4}
- C** 1.25×10^{-6}
- D** 4.8×10^{-5}

Question

The pH of 0.10 N acetic acid having $K_a = 1.8 \times 10^{-5}$ is

A 2.9

B 5.6

C 6.8

D 3.4

Question

A 0.01 M ammonia solution is 5% ionised. The concentration of OH ion is:

- A** 0.005 M
- B** 0.0001 M
- C** 0.0005 M
- D** 0.05 M

Question

The pH 0.005 M codeine ($\text{C}_{18}\text{H}_{21}\text{NO}_3$) solution is 9.95. Its pK_b value is:

- A** 8.92
- B** 5.80
- C** 3.76
- D** 4.29

Question

What is H^+ ion concentration of $5 \times 10^{-3} \text{ M}$ H_2CO_3 solution having a 10% dissociation

- A** 10^{-3}
- B** 10^{-2}
- C** 10^{-1}
- D** 5×10^{-2}

Question

The pH of a 0.1 M formic acid, which is 0.1% dissociated is equal to 4. What will be the pH of another weak monobasic acid (same concentration) which is 1% dissociated?

- A** 2
- B** 3
- C** 1
- D** 4

Question

At infinite dilution, the percentage ionisation for both strong and weak electrolytes is:

- A** 1%
- B** 20%
- C** 50%
- D** 100%

Question

0.01 M acetic acid solution is 1% ionised, then pH of this acetic acid solution is:

A 3

B 2

C 4

D 1

Question

pH of a strong diprotic acid (H_2A) at concentrations:

(i) 10^{-4} M

(ii) 10^{-4} N

are respectively:

A 3.7 and 4.0

B 4 and 3.7

C 4 and 4

D 3.7 and 3.7

Question

The neutralisation occurs when 10 mL of 0.1 M acid 'A' is allowed to react with 30 mL of 0.05 M base $M(OH)_2$. The basicity of the acid 'A' is_____.
[M is a metal]

Question

A weak acid HA has degree of dissociation x . Which option gives the correct expression of $(\text{pH} - \text{pK}_a)$?

A $\log (1 + 2x)$

B 0

C $\log \left(\frac{x}{1 - x} \right)$

D $\log \left(\frac{1 - x}{x} \right)$

Question

The pH of a 0.1 M aqueous solution of a weak acid (H_A) is 3. What is its degree of dissociation?

- A** 1%
- B** 10%
- C** 50%
- D** 25%

Question

For 10^{-3} M H_3PO_3 if $\alpha = 10\%$ then find out the value of pH?

A 3.7

B 4.7

C 5

D 2

Question

Find out K_a for 10^{-2} M HCN acid, having pOH 10:

A $K_a = 10^{-4}$

B $K_a = 10^{-2}$

C $K_a = 10^{-5}$

D $K_a = 10^{-6}$

Question

The pH of 0.15 M solution of HOCl ($K_a = 9.6 \times 10^{-6}$) is-

- A** 4.42
- B** 2.92
- C** 3.42
- D** None

Question

The K_a for a weak acid HX is 10^{-6} , then the ionisation constant of X^- will be:

- A** 10^{-6}
- B** 10^{-8}
- C** 10^{-14}
- D** 10^{-4}

Question

The $\text{p}K_{\text{b}}$ for fluoride ion at 25°C is 10.83, the ionisation constant of hydrofluoric acid at this temperature is:

- A** 2.72×10^{-5}
- B** 3.52×10^{-3}
- C** 6.76×10^{-4}
- D** 5.38×10^{-2}

Question

Determine the ionization constant of CH_3COOH if K_b for CH_3COO^- at 25°C is 5×10^{-10} .

Question

Calculate the pK_b of 0.004M Hydrazine if pH of the solution is 9.7. Also calculate K_a of $\text{NH}_2 - \text{NH}_3^\oplus$.

Question

Assertion: $\text{NaCl} + \text{HCl}$ does not experience common ion effect.

Reason: Both NaCl and HCl are strong electrolytes.

- A** Both A and R are correct and R is the correct explanation of A.
- B** Both A and R are correct, but R is not the correct explanation of A.
- C** A is correct but, R is incorrect.
- D** R is correct but, A is incorrect.

Question

The dissociation constant of acetic acid at a given temperature is 1.69×10^{-5} . The degree of dissociation of 0.01 M acetic acid in the presence of 0.01 M HCl is equal to

- A** 0.41×10^{-2}
- B** 0.18×10^{-7}
- C** 0.169×10^{-2}
- D** 0.013×10^{-4}

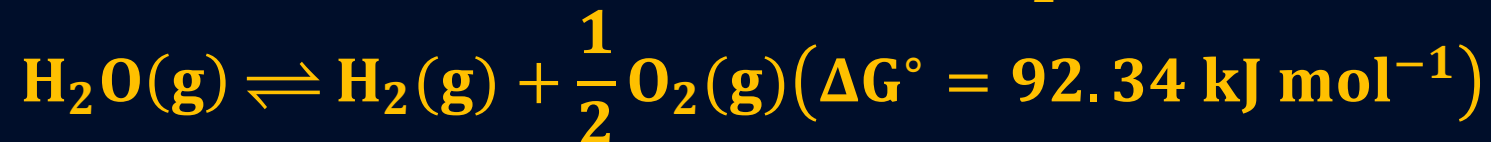
Question

Which of the following combinations of $\text{CH}_3\text{COOH} + \text{NaOH}$ would result in the formation of acidic buffer?

- A** 1 : 1 mol ratio
- B** 2 : 1 mol ratio
- C** 1 : 2 mol ratio
- D** Both (A) and (C)

Question

The equilibrium constant for decomposition of $\text{H}_2\text{O}(\text{g})$



is 8.0×10^{-3} at 2300 K and total pressure at equilibrium is 1 bar. Under this condition, the degree of dissociation (α) of water is _____ $\times 10^{-2}$ (nearest integer value)

[Assume a is negligible with respect to 1]

Question

For the given hypothetical reactions, the equilibrium constants are as follows:



The equilibrium constant for the reaction $X \rightleftharpoons W$ is

A 6.0

B 12.0

C 8.0

D 7.0

Question

The equilibrium constant for the reaction



is $K_c = 4.9 \times 10^{-2}$. The value of K_c for the reaction given below is



A 4.9

B 41.6

C 49

D 416

Question



For the given reaction, if the initial pressure is 450 mm Hg and the pressure at time t is 720 mm Hg at a constant temperature T and constant volume V . The fraction of A(g) decomposed under these conditions is $x \times 10^{-1}$. The value of x is ____ (nearest integer)

Question

A mixture of 1 mole of H_2O and 1 mole of CO is taken in a 10 litre container and heated to 725 K. At equilibrium 40% of water by mass reacts with carbon monoxide according to the equation :



The equilibrium constant $K_c \times 10^2$ for the reaction is _____.
(Nearest integer)

Question

The equilibrium composition for the reaction $\text{PCl}_3 + \text{Cl}_2 \rightleftharpoons \text{PCl}_5$ at 298 K is given below.

$$[\text{PCl}_3]_{\text{eq}} = 0.2 \text{ mol L}^{-1}$$

$$[\text{Cl}_2]_{\text{eq}} = 0.1 \text{ mol L}^{-1}$$

$$[\text{PCl}_5]_{\text{eq}} = 0.40 \text{ mol L}^{-1}$$

If 0.2 mol of Cl_2 is added at the same temperature, the equilibrium concentrations of PCl_3 is _____ $\times 10^{-2} \text{ mol L}^{-1}$.

Given: K_c for the reaction at 298 K is 20

Question

Consider the following equation: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H = -190 \text{ kJ}$

The number of factors which will increase the yield of SO_3 at equilibrium from the following is _____

- A. Increasing temperature
- B. Increasing pressure
- C. Adding more SO_2
- D. Adding more O_2
- E. Addition of catalyst

Question

Find out the value of α of 10^{-2} M NH_4OH solution if $[\text{OH}^-] = 10^{-3}$.

A 0.1%

B 10%

C 1%

D 5%

Question

For 10^{-3} M CH_3COOH solution if $K_a = 10^{-5}$ then pH is:

- A** 12
- B** 14
- C** 4
- D** 1

Question

What would be $[H^+]$ of 0.006 M benzoic acid ($K_a = 6 \times 10^{-5}$)

A 0.6×10^{-4}

B 6×10^{-4}

C 6×10^{-3}

D 3.6×10^{-4}

Question

The molarity of nitrous acid at which its pH becomes 2. ($K_a = 4.5 \times 10^{-4}$)-

A 0.3333

B 0.4444

C 0.6666

D 0.2222

Question

The ionisation constant of a tribasic acid is K_a . If its first, second and third ionisation constant are K_{a_1} , K_{a_2} and K_{a_3} respectively then-

A $K_a = K_{a_1} \times K_{a_2} \times K_{a_3}$

B $K_a = \frac{K_{a_1}}{K_{a_2} \times K_{a_3}}$

C $K_{a_2} = \frac{K_{a_1} \times K_a}{K_{a_3}}$

D None of these

Question

A buffer solution is one which has–

- A** reserved acid
- B** reserved base
- C** almost constant pH
- D** pH equal to 7

Question

The pH of blood circulating in a human body is maintained around 7.4 by the action of the buffer system:

- A** $\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}$
- B** $\text{NH}_4\text{Cl}/\text{NH}_3$
- C** $\text{H}_2\text{PO}_4^{2-}$
- D** $\text{H}_2\text{CO}_3/\text{HCO}_3^-$

Question

Buffer solutions can be prepared from mixtures of

- A** HCl and NaCl
- B** NaH_2PO_4 and Na_2HPO_4
- C** $\text{CH}_3\text{COOH} + \text{NaCl}$
- D** $\text{NH}_4\text{OH} + \text{NH}_3$

Question

Which one of the following pairs of solution is not an acidic buffer? [2015]

- A** H_2CO_3 and Na_2CO_3
- B** H_2PO_4 and Na_3PO_4
- C** HClO_4 and NaClO_4
- D** CH_3COOH and CH_3COONa

Question

Which will make basic buffer?

[2019]

- A** 100 mL of 0.1 M CH_3COOH + 100 mL of 0.1 M NaOH
- B** 100 mL of 0.1 M HCl + 200 mL of 0.1 M NH_4OH
- C** 100 mL of 0.1 M HCl + 100 mL of 0.1 M NaOH
- D** 50 mL of 0.1 M NaOH + 25 mL of 0.1 M CH_3COOH

Question

When 2 moles of HCl is added to 1 L. of an acidic buffer solution, its pH changes from 3.9 to 3.4. Find its buffer capacity.

Question

The pH of the solution containing 50mL, each of 0.10 M sodium acetate and 0.01 M acetic acid is

[Given pK_a of $CH_3COOH = 4.57$]

[2022]

A 2.57

B 5.57

C 3.57

D 4.57

Question

The pH of a buffer solution will be equal to the pK_a of acid, only if

- A** Molar concentration of acid is greater than molar concentration of conjugate base.
- B** Molar concentration of acid is smaller than molar concentration of conjugate base.
- C** Molar concentration of acid is equal to the molar concentration of conjugate base.
- D** Molar concentration of acid is almost negligible.

Question

In a mixture of a weak acid and its salt, the ratio of concentration of acid to salt is increased ten-fold. The pH of the solution

- A** Decreases by one
- B** Increases by one-tenth
- C** Increases by one
- D** Increases ten-fold

Question

The pK_a of a weak acid (HA) is 4.5. The pOH of an aqueous buffered solution of HA in which 50% of the acid is ionized is:

- A** 4.5
- B** 2.5
- C** 9.5
- D** 7.0

Question

20 mg of NaOH is added into CH_3COOH solution in which 60 mg of CH_3COOH is present. Calculate pH of solution if volume of final solution is 10 litre?
[Given $(\text{pK}_a)\text{CH}_3\text{COOH} = 4.74$]

Question

A certain buffer solution contains equal concentration of X^- and HX . The K_b for X^- is 1×10^{-10} . The pH of the buffer is–

- A** 4
- B** 7
- C** 10
- D** 14

Question

The ionization constant of dimethylamine is 5.4×10^{-4} . Calculate its degree of ionization in its 0.02M solution. What percentage of dimethylamine is ionized if the solution is also 0.1M in NaOH?

Question

The ionization constant of phenol is 1.0×10^{-10} . What is the concentration of phenolate ion in 0.05 M solution of phenol? What will be its degree of ionization if the solution is also 0.01M in sodium phenolate?

Question

Calculate the pH of a 0.10M ammonia solution. Calculate the pH after 50.0 mL of this solution is treated with 25.0 mL of 0.10M HCl. The dissociation constant of ammonia, $K_b = 1.77 \times 10^{-5}$.

Question

4.5 moles each of hydrogen and iodine is heated in a sealed ten litre vessel. At equilibrium, 3 moles of HI were found. The equilibrium constant for $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ is

Question

Water decomposes at 2300 K



**The percent of water decomposing at 2300 K and 1 bar is ____ (Nearest integer).
Equilibrium constant for the reaction is 2×10^{-3} at 2300K**

Question

37.8 g N_2O_5 was taken in a 1 L reaction vessel and allowed to undergo the following reaction at 500 K



The total pressure at equilibrium was found to be 18.65 bar

Then, $K_p = \underline{\hspace{2cm}} \times 10^{-2}$ [nearest integer]

Assume N_2O_5 to behave ideally under these conditions.

Given: $R = 0.082 \text{ bar L mol}^{-1} \text{ K}^{-1}$

Question

At 1990 K and 1 atm pressure, there are equal number of Cl_2 molecules and Cl atoms in the reaction mixture. The value of K_p for the reaction $\text{Cl}_2(\text{g}) \rightleftharpoons 2\text{Cl}(\text{g})$ under the above conditions is $x \times 10^{-1}$. The value of x is ____ (Rounded off to the nearest integer)

Question

If 40 ml of 0.2 M KOH is added to 160 ml of 0.1 M HCOOH [$K_a = 2 \times 10^{-4}$], the pH of the resulting solution is:

- A** 3.4
- B** 3.7
- C** 7
- D** 10.3

Question

To 1.0 L solution containing 0.1 mol each of NH_3 and NH_4Cl , 0.05 mol HCl is added. The change in pOH will be (pK_a for $\text{NH}_3 = 4.74$)

- A** 0.30
- B** -0.30
- C** 0.48
- D** -0.48

Question

Solution of 0.1 M NH_4OH and 0.1 M NH_4Cl has pH 9.25, Then find out pK_b of NH_4OH

- A** 9.25
- B** 4.75
- C** 3.75
- D** 8.25

Question

100 mL of a buffer solution contains 0.1M each of weak acid HA and salt NaA. How many gram of NaOH should be added to the buffer so that its pH will be 6 (K_a of HA = 10⁻⁵)

- A** 0.328
- B** 0.458
- C** 4.19
- D** None

Question

Which salt is not an example of acidic salt:

- A** HCOONa
- B** NaH_2PO_2
- C** NaHS
- D** (A) and (B) Both

Question

A homogeneous ideal gaseous reaction $\text{AB}_{2(g)} \rightleftharpoons \text{A}_{(g)} + 2\text{B}_{(g)}$ is carried out in a 25 litre flask at 27°C . The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm . The value of K_p is $x \times 10^{-2}$. The value of x is _____ (Integer answer). $[R = 0.08206 \text{ dm}^3 \text{ atm K mol}^{-1}]$

Question

In a chemical reaction $A + 2B \xrightleftharpoons{K} 2C + D$, the initial concentration of B was 1.5 times of A but the equilibrium concentrations of A and B were found to be equal. The equilibrium, constant (K) for the aforesaid chemical reaction is

A 4

B 16

C $1/4$

D 1

Question

Consider the reaction $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$

The equilibrium constant of the above reaction is K_p . If pure ammonia is left to dissociate, the partial pressure of ammonia at equilibrium is given by (Assume that $P_{\text{NH}_3} \ll P_{\text{Total}}$ at equilibrium)

A $\frac{3^{\frac{3}{2}} K_p^{\frac{1}{2}} P^2}{16}$

B $\frac{K_p^{\frac{1}{2}} P^2}{16}$

C $\frac{K_p^{\frac{1}{2}} P^2}{4}$

D $\frac{3^{\frac{3}{2}} K_p^{\frac{1}{2}} P^2}{4}$

Question

An aqueous solution of sodium carbonate is alkaline because sodium carbonate is a salt of

- A** Weak acid and weak base
- B** Strong acid and weak base
- C** Weak acid and strong base
- D** Strong acid and strong base

Question

Which of the following cannot be hydrolysed

- A** A salt of weak acid and strong base
- B** A salt of strong acid and weak base
- C** A salt of weak acid and weak base
- D** A salt of strong acid and strong base

Question

Assertion: An aqueous solution of NH_4NO_3 is acidic in characters.

Reason: NH_4NO_3 in an aqueous solution undergoes anionic hydrolysis.

- A** Both A and R are correct and R is the correct explanation of A.
- B** Both A and R are correct, but R is not the correct explanation of A.
- C** A is correct but, R is incorrect.
- D** R is correct but, A is incorrect.

Question

Which is the correct alternate for hydrolysis constant of NH_4CN

- A** $\sqrt{\frac{K_w}{K_a}}$
- B** $\frac{K_w}{K_a \times K_b}$
- C** $\sqrt{\frac{K_b}{c}}$
- D** $\frac{K_a}{K_b}$

Question

If a salt of strong acid and weak base hydrolyses appreciably ($\alpha = 0.1$), which of the following formula is to be used to calculate degree of hydrolysis ' α '?

A $\alpha = \sqrt{\frac{K_w}{K_a \cdot a}}$

B $\alpha = \sqrt{\frac{K_w}{K_b \cdot K}}$

C $\alpha = \sqrt{\frac{K_w}{K_b \cdot Kb}}$

D None of these

Question

Calculate pH of 10^{-2} M KCN solution if dissociation constant of HCN = 10^{-10} .

Question

Calculate the degree of hydrolysis of a mixture containing 0.1N NH_4OH and 0.1 N HCN If $K_a = 10^{-5}$ and $K_b = 10^{-5}$

Question

What is the pH of 1 M CH_3COONa solution? K_a of acetic acid = 1.8×10^{-5}

$K_w = 10^{-14} \text{ mol}^2 \text{ L}^{-2}$

- A** 2.4
- B** 3.6
- C** 4.8
- D** 9.4

Question

Calculate the pH of aqueous solution of 1.0 M HCOONH_4 assuming complete dissociation (pK_a of $\text{HCOOH} = 3.8$ and pK_b of $\text{NH}_3 = 4.8$)

A 3.5

B 4.5

C 5.5

D 6.5

Question

pK_a of a weak acid (HA) and pK_b of a weak base (BOH) are 3.2 and 3.4, respectively. The pH of their salt (AB) solution is:

- A** 6.9
- B** 7.0
- C** 1.0
- D** 7.2

Question

pH of 0.1 M NaCN solution is 11, then the percentage hydrolysis is–

- A** 0.1%
- B** 1%
- C** 0.01%
- D** 10%

Question

HX is a weak acid ($K_a = 10^{-5}$). It forms a salt NaX (0.1 M) on reacting with caustic soda. The degree of hydrolysis of NaX is:

- A** 0.01%
- B** 0.0001%
- C** 0.1%
- D** 0.5%

Question

The pK_b of dimethylamine and pK_a of acetic acid are 3.27 and 4.77 respectively at T(K). The correct option for the pH of dimethylammonium acetate solution is:

- A** 7.75
- B** 6.25
- C** 8.50
- D** None of these

Question

The pH of 0.1 M NH_4Cl solution is 5.13. What will be the dissociation constant of NH_4OH

- A** 1.8×10^{-7}
- B** 1.8×10^{-9}
- C** 1.8×10^{-5}
- D** None

Question

**What is the hydrolysis constant of the OCl^- ion?
The ionization constant of HOCl is 3.0×10^{-8} .**

- A** 3.33×10^{-8}
- B** 3.33×10^{-7}
- C** 3.0×10^{-7}
- D** 3.33×10^{-6}

Question

Let the solubility of an aqueous solution of $\text{Mg}(\text{OH})_2$ be x then its K_{sp} is

- A** $4x^3$
- B** $108x^5$
- C** $27x^4$
- D** $9x$

Question

A salt M_2X_3 dissolves in water such that its solubility is x mole/litre. Its K_{sp} is:

- A** x^5
- B** $6x^2$
- C** $108 x^5$
- D** $6x^5$

Question

The K_{sp} for $\text{Cr}(\text{OH})_3$ is 1.6×10^{-30} . The molar solubility of this compound in water is

A $\sqrt[2]{1.6 \times 10^{-30}}$

B $\sqrt[4]{1.6 \times 10^{-30}}$

C $\sqrt[4]{\frac{1.6 \times 10^{-30}}{27}}$

D $\frac{1.6 \times 10^{-30}}{27}$

Question

Which is the correct representation of the solubility product constant of Ag_2CrO_4

- A** $[\text{Ag}^+]^2 [\text{CrO}_4^{-2}]$
- B** $[\text{Ag}^+] [\text{CrO}_4^{-2}]$
- C** $[2\text{Ag}^+] [\text{CrO}_4^{-2}]$
- D** $[2\text{Ag}^+]^2 [\text{CrO}_4^{-2}]$

Question

The solubility in water of a sparingly soluble salt AB_2 is $1.0 \times 10^{-5} \text{ mol L}^{-1}$. Its solubility product will be

- A** 4×10^{-15}
- B** 4×10^{-10}
- C** 1×10^{-15}
- D** 1×10^{-10}

Question

The solubility product of a salt having general formula MX_2 , in water is 4×10^{-12} . The concentration of M^{2+} ions in the aqueous solution of the salt is

A $2.0 \times 10^{-6} \text{ M}$

B $1.0 \times 10^{-4} \text{ M}$

C $1.6 \times 10^{-4} \text{ M}$

D $4.0 \times 10^{-10} \text{ M}$

Question

Zirconium phosphate $[\text{Zr}_3(\text{PO}_4)_4]$ dissociates into three zirconium cations of charge +4 and four phosphate anions of charge -3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by K_{sp} then which of the following relationship between S and K_{sp} is correct?

- A** $S = \{K_{\text{sp}}/(6912)^{1/7}\}$
- B** $S = (K_{\text{sp}}/6912)^{1/7}$
- C** $S = \{K_{\text{sp}}/144^{1/7}\}$
- D** $S = \{K_{\text{sp}}/(6912)^7\}$

Question

The necessary condition for saturated solution is–

- A** Product of ionic concentrations = Solubility product
- B** Product of ionic concentrations < solubility product
- C** Product of ionic concentrations > solubility product
- D** None of the above

Question

When 15 mL of 0.05 M AgNO_3 is mixed with 45.0 mL of 0.03 M K_2CrO_4 , predict whether precipitation of Ag_2CrO_4 occurs or not? K_{sp} of $\text{Ag}_2\text{CrO}_4 = 1.9 \times 10^{-12}$.

Question

A precipitate of AgCl is formed when equal volumes of the following are mixed
[K_{sp} for AgCl = 10^{-10}]

- A** 10^{-4} M AgNO₃ and 10^{-7} M HCl
- B** 10^{-5} M AgNO₃ and 10^{-7} M HCl
- C** 10^{-5} M AgNO₃ and 10^{-4} M HCl
- D** 10^{-6} M AgNO₃ and 10^{-6} M HCl

Question

When equal volumes of the following solutions are mixed, precipitation of Ag ($K_{sp} = 1.8 \times 10^{-10}$) will occur only with

- A** $10^{-4} \text{ M}(\text{Ag}^+)$ and $10^{-4} \text{ M}(\text{Cl}^-)$
- B** $10^{-5} \text{ M}(\text{Ag}^+)$ and $10^{-5} \text{ M}(\text{Cl}^-)$
- C** $10^{-6} \text{ M}(\text{Ag}^+)$ and $10^{-6} \text{ M}(\text{Cl}^-)$
- D** $10^{-10} \text{ M}(\text{Ag}^+)$ and $10^{-10} \text{ M}(\text{Cl}^-)$

Question

At 25°C, the solubility product of $\text{Mg}(\text{OH})_2$ is 1.0×10^{-11} . At which pH, will Mg^{2+} ions start precipitating in the form of $\text{Mg}(\text{OH})_2$ from a solution of 0.001 M Mg^{2+} ions?

A 9

B 10

C 11

D 8

Question

The solubility product of different sparingly soluble salts are

1. $XY = 4 \times 10^{-2}$

2. $X_2Y = 3.2 \times 10^{-11}$

3. $XY_3 = 2.7 \times 10^{-31}$

The increasing order of solubility is

A 1, 3, 2

B 2, 1, 3

C 1, 2, 3

D 3, 1, 2

Question

The K_{sp} of AgCl at 25°C is 1.5×10^{-10} . Find the solubility of AgCl (in gm/L) in an aqueous solution containing 0.01 M AgNO₃.

Question

What is the molar solubility of Al(OH)_3 in 0.2 M NaOH solution? Given that, solubility product of $\text{Al(OH)}_3 = 2.4 \times 10^{-24}$:

- A** 12×10^{-23}
- B** 12×10^{-21}
- C** 3×10^{-19}
- D** 3×10^{-22}

Question

If K_{sp} of Ag_2CO_3 is 8×10^{-12} , the molar solubility of Ag_2CO_3 in 0.1 M AgNO_3 is:

- A** $8 \times 10^{-12} \text{ M}$
- B** $8 \times 10^{-11} \text{ M}$
- C** $8 \times 10^{-10} \text{ M}$
- D** $8 \times 10^{-13} \text{ M}$

Question

Calculate the simultaneous solutions of AB and AC when as $(K_{sp})_{AB} = 10^{-12}$ and $(K_{sp})_{AC} = 5 \times 10^{-13}$.

Question

CaCO_3 and BaCO_3 have solubility product values 1×10^{-8} and 5×10^{-9} , respectively. If water is shaken up with both solids till equilibrium is reached, the concentration of CO_3^{2-} ion is:

- A** 1.5×10^{-8}
- B** 1.225×10^{-4}
- C** 2.25×10^{-9}
- D** None of these

Question

Incorrect statement for the use of indicators in acid-base titration is:

- A** Methyl orange may be used for a weak acid vs weak base titration.
- B** Methyl orange is a suitable indicator for a strong acid vs weak base titration
- C** Phenolphthalein is a suitable indicator for a weak acid vs strong base titration
- D** Phenolphthalein may be used for a strong acid vs strong base titration.

Question

Given below are two statements:

Statement-I: On passing $\text{HCl}_{(g)}$ through a saturated solution of BaCl_2 , at room temperature white turbidity appears.

Statement-II: When HCl gas is passed through a saturated solution of NaCl , sodium chloride is precipitated due to common ion effect.

In the light of the above statements, choose the most appropriate answer from the options given below:

- A** Statement-I is correct but Statement-II is incorrect
- B** Both Statement-I and Statement-II are incorrect
- C** Statement-I is incorrect but Statement-II is correct
- D** Both Statement-I and Statement-II are correct

Question

An indicator 'X' is used for studying the effect of variation in concentration of iodide on the rate of reaction of iodide ion with H_2O_2 at room temp. The indicator 'X' forms blue coloured complex with compound 'A' present in the solution. The indicator 'X' and compound 'A' respectively are

- A** Starch and iodine
- B** Methyl orange and H_2O_2
- C** Starch and H_2O_2
- D** Methyl orange and iodine

Question

Which of the following happens when NH_4OH is added gradually to the solution containing 1M A^{2+} and 1M B^{3+} ions?

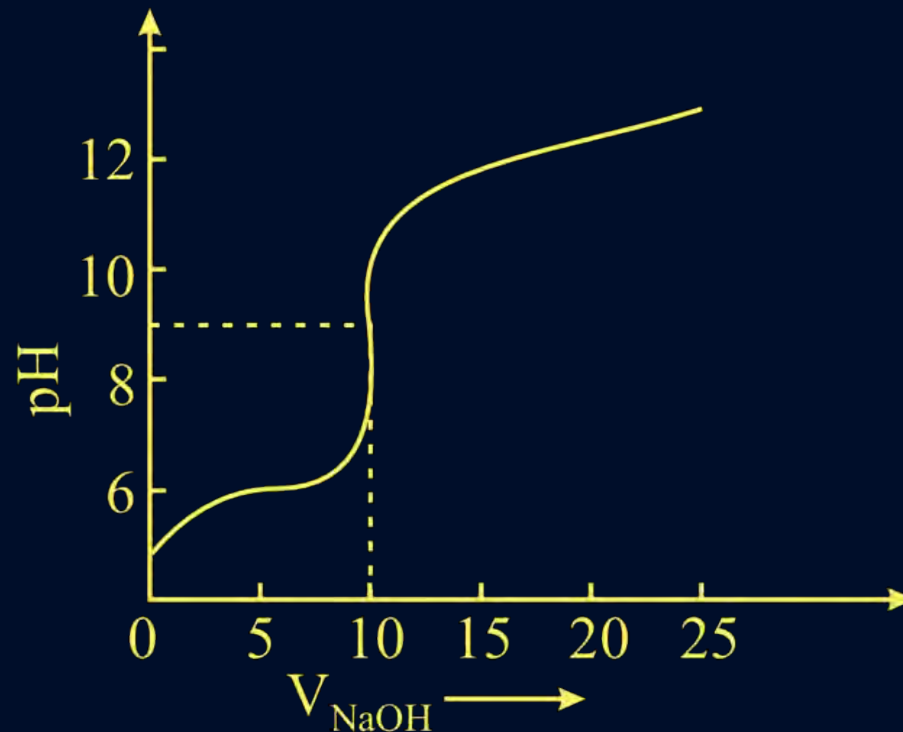
Given: $K_{\text{sp}}[\text{A}(\text{OH})_2] = 9 \times 10^{-10}$ and $K_{\text{sp}}[\text{B}(\text{OH})_3] = 27 \times 10^{-18}$ at 298 K .

- A** Both $\text{A}(\text{OH})_2$ and $\text{B}(\text{OH})_3$ do not show precipitation with NH_4OH
- B** $\text{B}(\text{OH})_3$ will precipitate before $\text{A}(\text{OH})_2$
- C** $\text{A}(\text{OH})_2$ will precipitate before $\text{B}(\text{OH})_3$
- D** $\text{A}(\text{OH})_2$ and $\text{B}(\text{OH})_3$ will precipitate together

Question

The titration curve of weak acid vs. strong base with phenolphthalein as indicator) is shown below. The $K_{\text{phenolphthalein}} = 4 \times 10^{-10}$.

Given: $\log 2 = 0.3$



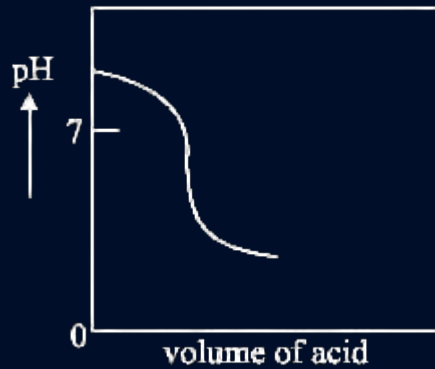
The number of following statements which is/are correct about phenolphthalein is _____ [8 April, 2023 (Shift-I)]

- A** It can be used as an indicator for the titration of weak acid with weak base.
- B** It begins to change colour at $\text{pH} = 8.4$
- C** It is a weak organic base
- D** It is colourless in acidic medium

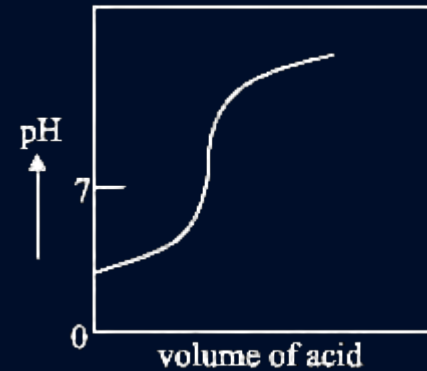
Question

The plot of pH-metric titration of weak base NH_4OH vs strong acid HCl looks like: [27 July, 2022 (Shift-II)]

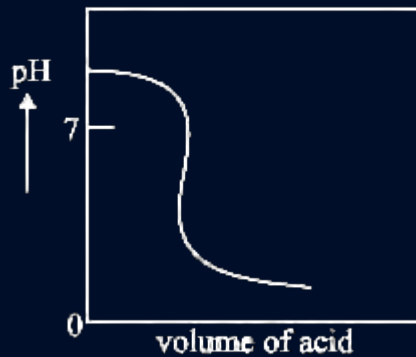
A



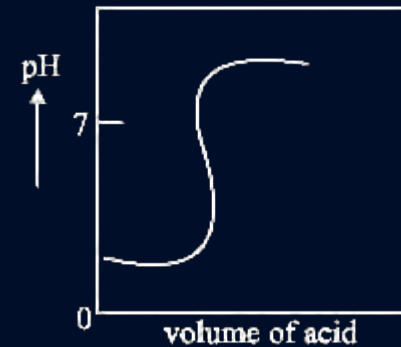
B



C



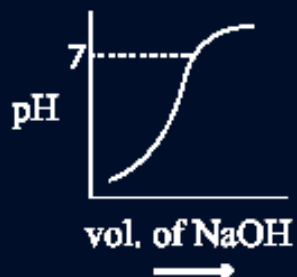
D



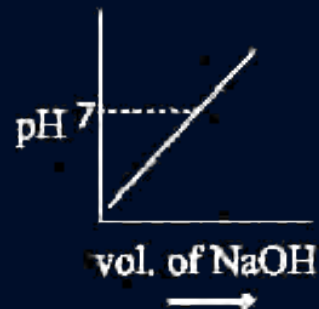
Question

100 mL of 0.1 M HCl is taken in a beaker and to it 100 mL of 0.1 M NaOH is added in steps of 2 mL and the pH is continuously measured. Which of the following graphs correctly depicts the change in pH? [3 Sept, 2020 (Shift-II)]

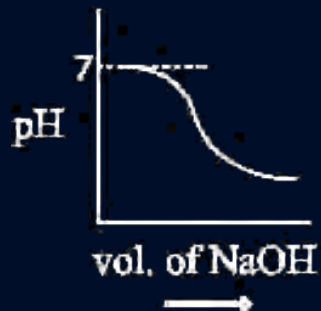
A



B



C



D

